

Discrete Mathematics and Its Applications

Introductory Lecture

What is Discrete Mathematics?

Discrete mathematics is the part of mathematics devoted to the study of discrete (as opposed to continuous) objects.

Calculus deals with continuous objects and is not part of discrete mathematics.

Examples of discrete objects: integers, steps taken by a computer program.

A course in discrete mathematics provides the mathematical background needed for all subsequent courses in computer science.

Kinds of Problems Solved Using Discrete Mathematics₁

How many ways can a password be chosen following specific rules?

How many valid Internet addresses are there?

What is the probability of winning a particular lottery?

Is there a link between two computers in a network?

How can I encrypt a message so that no unintended recipient can read it?

Kinds of Problems Solved Using Discrete Mathematics₂

Find the shortest tour that visits each of a group of cities only once and then ends in the starting city.

How can a list of integers be sorted so that the integers are in increasing order?

How many steps are required to do such a sorting?

Goals of a Course in Discrete Mathematics₁

Mathematical Reasoning: Ability to read, understand, and construct mathematical arguments and proofs.

Combinatorial Analysis: Techniques for counting objects of different kinds.

Discrete Structures: Abstract mathematical structures that represent objects and the relationships between them. Examples are sets, permutations, relations, graphs, trees.

Goals of a Course in Discrete Mathematics₂

Algorithmic Thinking: One way to solve many problems is to specify an algorithm. An algorithm is a sequence of steps that can be followed to solve any instance of a particular problem.

Applications and Modeling: It is important to appreciate and understand the wide range of applications of the topics in discrete mathematics and develop the ability to develop new models in various domains.

Discrete Mathematics is a Gateway Course

Topics in discrete mathematics will be important in many courses that you will take in the future:

- **Computer Science:** Computer Architecture, Data Structures, Algorithms, Programming Languages, Compilers, Computer Security, Databases, Artificial Intelligence, Networking, Graphics, Game Design, Theory of Computation,